

Question	Answer	Marks
11(a)	no forbidden band / valence and conduction bands overlap	B1
	no change in number of charge carriers (as temperature rises)	B1
	increased lattice vibrations so resistance increases	B1
11(b)	photons captured / absorbed by electrons in valence band	B1
	electrons promoted to conduction band	B1
	leaving holes in the valence band	B1
	more holes and / or electrons so resistance decreases	B1

Question	Answer	Marks
12(a)	Any 2 from:	B2